

Online PAC Labeling from a Linear Programming Perspective

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Recap: AI-assisted labeling

We have a pool of unlabeled items $X_1, \dots, X_T$. Need to collect their true labels $Y_1, \dots, Y_T$.

Two ways to get a label $\tilde{Y}_t$:

- **ML models** $\hat{Y}_t^{(a)}$ — sometimes *wrong*, but *cheap*
- **Expert** Y_t — *correct*, but *expensive*.

Two Goals:

- Save cost by reducing the number of expert queries.
- Probably Approximately Correct (PAC) guarantee:

$$\frac{1}{T} \sum_{t=1}^T \ell(Y_t, \tilde{Y}_t) \leq \varepsilon \quad w.p. \geq 1 - \alpha.$$

Problem setup

- For each item X_t , decide how to label it: use the **expert** or one of k **cheap models**.
- Choose action a from an action set $\mathcal{A} = \{E, 1, \dots, k\}$.
- For item t and action a , a labeling **loss** and a **cost**:

$$h_t(a) = \begin{cases} 0, & a = E \\ \ell(Y_t, \hat{Y}_t^{(a)}), & a \in [k] \end{cases} \quad c_t(a) = \begin{cases} 1, & a = E \\ 0, & a \in [k] \end{cases}$$

- Released label $\tilde{Y}_t = Y_t$ if we defer, else $\hat{Y}_t^{(a_t)}$.

Key quantities: *cost* we want to minimize, *error* we must control.

The cost-aware labeling LP

Let $x_{t,a} \in [0, 1]$ be the probability of taking action a on item t .

$$\begin{aligned} \min_x \quad & \underbrace{\frac{1}{T} \sum_t \sum_a c_t(a) x_{t,a}}_{\text{average cost}} \\ \text{s.t.} \quad & \underbrace{\frac{1}{T} \sum_t \sum_a h_t(a) x_{t,a}}_{\text{average error budget}} \leq \varepsilon, \quad \sum_a x_{t,a} = 1, \quad x_{t,a} \geq 0. \end{aligned}$$

- **One** global constraint: keep average error $\leq \varepsilon$.
- Naturally handles *multiple* cheap sources and item-specific costs.
- If true losses $h_t(a)$ were known, this LP is the *optimal* policy.

The dual gives a single “price of error”

One Lagrange multiplier $\lambda \geq 0$ for the error constraint. The dual:

$$\max_{\lambda \geq 0} -\lambda \varepsilon + \frac{1}{T} \sum_{t=1}^T \min_{a \in \mathcal{A}} \{c_t(a) + \lambda h_t(a)\}.$$

The fixed-price rule

Once λ is fixed, the decision **separates across items**:

$$a_t(\lambda) \in \arg \min_{a \in \mathcal{A}} \{c_t(a) + \lambda h_t(a)\}.$$

λ is the exchange rate between cost and error.

Small $\lambda \Rightarrow$ trust AI | Large $\lambda \Rightarrow$ defer to expert.

What the price actually does: a threshold

With equal cheap-source costs, first **route** to the best cheap model

$$j_t^* \in \arg \min_{j \in [k]} \hat{h}_t(j), \quad s_t = \min_j \hat{h}_t(j) \quad (\text{the “uncertainty score”}).$$

Then λ only decides *accept vs. defer* via a threshold $1/\lambda$:

$$s_t \geq 1/\lambda \Rightarrow \text{defer to expert}$$

$$s_t < 1/\lambda \Rightarrow \text{accept cheap label.}$$

- Recovers the original **PAC-labeling** threshold rule (single source: $\hat{h}_t = \text{uncertainty}$).
- Multi-source extension is “free”: just replace the score by $\min_j \hat{h}_t(j)$.
- Two regimes ahead: **offline** (fix λ once) and **online** (move λ_t over time).

Part I — Offline PAC Labeling

Certify one price on a calibration set, then label the whole pool.

Offline: certify a price from a small labeled sample

Idea. The cost $C_T(\lambda)$ is known from proxies; only the error $L_T(\lambda)$ needs true labels. Estimate it on a calibration set.

- Sample m items, get expert labels, form empirical loss $\hat{L}_m(\lambda)$.
- Hoeffding upper confidence bound:

$$U_m(\lambda; \alpha) = \hat{L}_m(\lambda) + \sqrt{\frac{\log(1/\alpha)}{2m}}.$$

- Pick the **cheapest certified price**:

$$\hat{\lambda} = \min\{\lambda \geq 0 : U_m(\lambda; \alpha) \leq \varepsilon\}.$$

Why no union bound over λ ?

As λ grows, items only move *AI* $\rightarrow$ *expert*. So $L_T(\lambda)$ is **monotone** — a single nested family, certified for free.

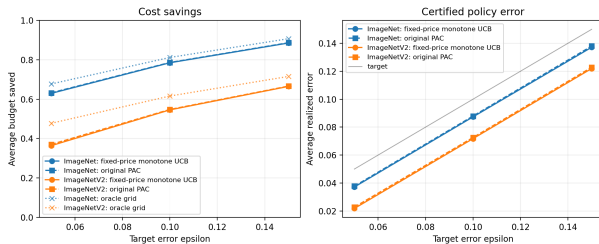
Theorem (offline monotone-threshold PAC)

With probability at least $1 - \alpha$ over the calibration draw, the released labels satisfy

$$\frac{1}{T} \sum_{t=1}^T \ell(Y_t, \tilde{Y}_t) \leq \varepsilon.$$

- Distribution-free, finite-sample. Falls back to expert-only if no finite price qualifies.
- **Corollary:** on a fresh deployment pool of size n , error $\leq \varepsilon + \sqrt{\log(1/\delta)/2n}$ w.p. $1 - \alpha - \delta$.
- Proxies affect *cost*, not *validity*: a bad proxy just defers more often.

Offline experiments: LP rule matches original PAC



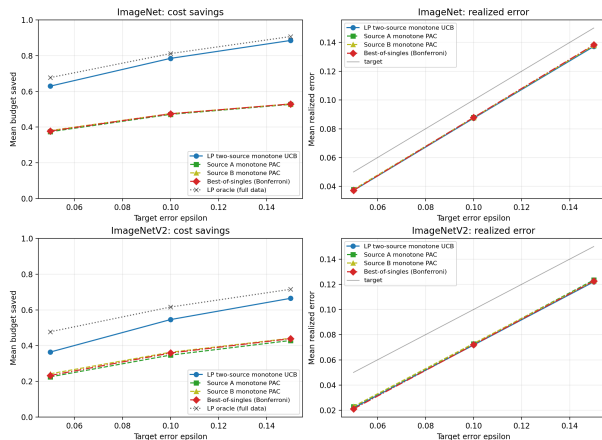
One cheap source, ImageNet / ImageNetV2.

One source ($\varepsilon = 0.10$):

Data	Method	Error	Saved
IN	LP	0.087	0.785
IN	PAC	0.088	0.786
INv2	LP	0.072	0.546
INv2	PAC	0.073	0.548

- Error stays under target ε (0 violations).
- LP fixed-price $\approx$ original PAC, as predicted.
- Saves $\sim 78\%$ of expert budget on ImageNet.

Offline experiments: routing beats single sources



Two sources ($\varepsilon = 0.10$, ImageNet):

Method	Error	Saved
LP routing	0.087	0.785
Best-of-singles	0.088	0.475

- Per-item routing to the lower-proxy source.
- Same error control, but **much** larger budget savings (0.78 vs 0.47).
- This is where the LP view earns its keep.

Two complementary cheap sources (by class parity).

Part II — Online PAC Labeling via Audits

Items stream in; labels are hidden unless we audit.

Online: a moving threshold + randomized audits

Items arrive one at a time. AI losses are *unobserved* unless we pay the expert.

- Keep a moving price λ_t ; route then threshold: $D_t = \mathbf{1}\{\hat{h}_t^\lambda \geq 1/\lambda_t\}$ (defer) .
- If we use the AI label, **audit** it with probability $p_t \in [p_{\min}, 1]$:

$$A_t \sim \text{Bernoulli}(p_t).$$

If audited, query the expert, *correct* the label, and observe the AI's true error.

Inverse-probability feedback

$$\hat{Z}_t = \frac{O_t}{q_t} (H_t - \varepsilon_0), \quad \mathbb{E}[\hat{Z}_t \mid \mathcal{G}_t] = H_t - \varepsilon_0 \text{ (unbiased).}$$

Audited rare errors are up-weighted by $1/p_t$ to stay unbiased.

Online: the price update

$$\lambda_{t+1} = [\lambda_t + \eta \hat{Z}_t]_+.$$

- An audited **mistake** pushes λ_t *up* $\Rightarrow$ threshold $1/\lambda_t$ drops $\Rightarrow$ routing gets more conservative.
- A **deferral** or a clean audit pushes λ_t *down* $\Rightarrow$ trust the AI more.
- Internal target $\varepsilon_0 \leq \varepsilon$ leaves slack for the finite-horizon bound.

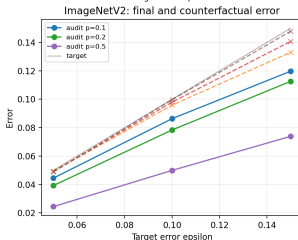
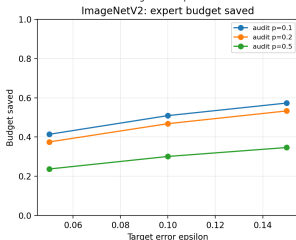
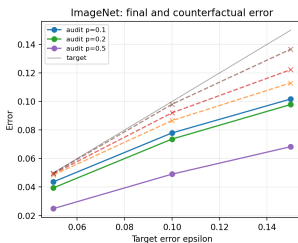
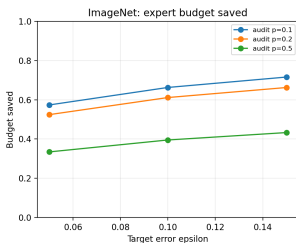
Theorem (audited online finite-horizon control)

With probability $\geq 1 - \delta$ over audit randomness, for *any* data sequence,

$$\frac{1}{T} \sum_{t=1}^T G_t \leq \varepsilon_0 + \frac{\Lambda_{\text{on}}}{\eta T} + \frac{1}{\rho_{\min}} \sqrt{\frac{2 \log(1/\delta)}{T}}.$$

Choosing $\eta \asymp T^{-1/2}$ gives final error $\leq \varepsilon$, slack $O(T^{-1/2})$.

Online experiments: the validity–cost knob



Online ($\varepsilon = 0.10$):

Data	p	Error	Saved
IN	0.1	0.078	0.663
IN	0.2	0.074	0.612
IN	0.5	0.049	0.395
INv2	0.2	0.078	0.468

- Smaller p : save more budget.
- Larger p : correct more, lower final error.
- Released-label error tracked below target under partial feedback.

Audited online sweep. Dashed = AI error before correction.

Takeaways

One idea, two regimes

- LP dual $\Rightarrow$ a single *price of error* λ .
- **Offline:** certify $\hat{\lambda}$ on held-out labels.
- **Online:** move λ_t with audited inverse-probability feedback.

What we showed

- Proxies affect *cost*, not *validity*.
- LP rule matches PAC with one source; *routing wins* with two.
- Online stays valid under partial feedback; p trades cost vs. error.

Limitation & next step

Current data has only one real cheap source. The interesting case is many sources with *heterogeneous costs*, where a scalar threshold no longer suffices.

Thank you! Questions?